

Interface Capability Brief

EDI 837 NCCI MUE · Stakeholder overview · 2026-08-13

1. Executive summary

Show that PilotFish can apply CMS Medically Unlikely Edits (MUE) as a business-rule layer on X12 837P claims: look up each service line's CPT/HCPCS against an external MUE table and kick out lines whose units exceed the max. There is no stock PilotFish NCCI/MUE module — this is the same catalog + XSLT pattern already used in Med Rec (MUE_EDITS.CPT / MAX_VALUE_PER_LINE), pointed at 837 instead of HL7 DFT.

2. Who this is for

- Sources: Directory-drop 837P (input/inbound/*.edi) - Destinations: output/pass/ (within MUE limit), output/kickout/ (units exceeded) - Demo vs production: Demo only — synthetic CMS-shaped table (not the official quarterly NCCI files). Production would load CMS MUE CSVs into SQL (or a file lookup) on the CMS publish cycle. - Sibling demo: edi-837-ncci-ntp (procedure-to-procedure pairs). Med Rec does not implement PTP.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format / envelope: X12 837P 005010X222A1, //Transaction fork (one ST per claim)
- Identity fields: CLM01 claim control #; first SV1 CPT (SV101-02) + units (SV104)
- Samples path: samples/sample_837p_mue.edi

Integration highlights

FQCN sources: edi-837p-qcare, edi-835-payment-integrity, Med Rec MUE_EDITS / DFT split XSLT, PilotFish Documentation 26R1.11. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	com.pilotfish.eip.modules.file.DirectoryListener	Poll \$EDI_INBOUND_DIRECTORY. Skeleton: edi-837p-qcare
R1 Format	EDITransformationModule + XPathForkingModule //Transaction	TableData edi-tabledata/837-Q1 (5010). Playbook §3.6. Docs 26R1.11 vs image 23R1.
R1 Target processors	XPath identity → XSLT line extract → DatabaseSqlProcessor MueEdits → XSLT decision	Catalog shape matches Med Rec MUE_EDITS (CPT + max units per line). SQL processor skeleton: edi-835-payment-integrity
R1 Transport	DirectoryTransport + XML Formatting on the transport	Stage under output/staged-decisions/ (playbook §1.4)
R2 Listener	DirectoryListener	Poll staged decisions

Stage	Module / mechanism	Notes
R2 Router	XPathRoutingModule accumulate=true	//MatchBucket = pass / kickout
R2 Transports	DirectoryTransport	Unique Pretty-Print names per target

4. Honest boundaries (not claimed yet)

- Load a real CMS MUE quarterly extract in a later pass?
- Split-to-limit (Med Rec DFT policy) as an optional pass-path rewrite?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll And Score MUE” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: EDI parses; first SV1 CPT looked up in MueEdits; billed units vs MaxUnits - What is NOT checked: SNIP 1–7, MAI 2/3 date-of-service aggregation, multi-line MUE (demo is one SV1 per ST), official CMS file ingest, PTP (see sibling demo) - Does failure block outbound? yes for unparseable EDI; kickout theater for MUE exceed

6. State & idempotency

- Status model: none (stateless file buckets). Catalog is read-only seed. - When state advances: after MueDecision is staged and routed - Dedup keys: CLM01 (overwrite same filename on resubmit) - Retry / poison: unparseable 837 fails at EDI→XML (fail-closed)

8. Dual-write / side effects

- Order: decision XML → route to pass or kickout file - Compensation: none - Demo shortcuts: one procedure line per claim; synthetic four-row catalog; pass-through of original 837 is not re-emitted (decision XML is the artifact)

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll And Score MUE: runs steps such as Poll EDI 837P Files, Fork - XPath, Stage MUE Decision.

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2 - Route MUE Decisions: runs steps such as Poll Staged MUE Decisions, Conditional Router, Write Pass, Write Kickout.

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7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No stock NCCI module — Product has SNIP, not CMS MUE/PTP tables
- Risk noted in design: Med — 23R1 trial X12 tables expired — Sandbox mounts EDI/TableData/x12
- Risk noted in design: Med — Official CMS quarterly files not loaded — Demo seed only
- Risk noted in design: Med — Duplicate processor names — EIP refuses the route
- Risk noted in design: Low — SQLXML tags UPPERCASE — //MaxUnits misses MAXUNITS

8. Seeing it work

- - Ports: SQL 14342, EIP 8130, Web UI 8131 - Volumes: ./input, ./output, ./logs, ./samples, ./documents, TableData mount - Heap: 512M–2GB - Dependencies / cold start: SQL health + seed ~30–60s, EIP ~60–90s - Credentials: sa / PilotFish_Demo1! · Edi837NcciMue
- Local: http://127.0.0.1:8131/
- EIP: http://127.0.0.1:8130/eip/

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- SQL: 127.0.0.1:14342 (sa / PilotFish_Demo1! · Edi837NcciMue)
 - docker compose --profile full up -d --build

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.